Safety in Numbers: How Weather Conditions Predict Olive Ridley Arribada Events

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Sea turtles are an iconic and charismatic family of marine reptiles widely recognized for their large shells. However, not all sea turtles have hardback shells. The leatherback sea turtle is the one living species of sea turtle belonging to the Dermochelyidae family that, as its name suggests, has leather-like skin covering small, bony plates across its back. However, the other six species of sea turtle belonging to the Cheloniidae family all have hard shells, consisting of a carapace along its back and a plastron along its underside (Bennett 2018). Other than their shells, sea turtles are also uniquely adapted to survive in the open ocean through traits like enhanced thermal regulation through blood circulation, the ability to hold their breath for extended periods of time, strong flippers to help them swim, and keratin spikes in their throats that help them feed underwater (Bennett 2018; Osterloff 2025).

Sea turtles play a crucial ecological role in coastal and marine environments. In general, they help to regulate the health of seagrass and coral communities by grazing and controlling populations of sponges and jellyfish (Bennett 2018). They are also economically and culturally significant, having been the focus of cultural traditions and mythology for centuries, including those of the Iroquois people in North America, the Moche people of Peru, the Seri Indians from the Gulf of California, the Ancient Mayans, and the Ancient Chinese (Bennett 2018).

Literature Review

The Olive Ridley sea turtle (*Lepidochelys olivacea*) is a dominantly pelagic species globally distributed across the tropical regions of the Pacific, Atlantic, and Indian oceans, and is the most abundant of the world's sea turtles (NOAA, 2025). *L. olivacea* belongs to the Cheloniidae family, one of the two sea turtle families, and its common name comes from its green color and heart-shaped shell (NOAA, 2025).

Though they live in the open ocean, adult Olive Ridleys periodically aggregate along the coast to reproduce. *L. olivacea* can engage in solitary nesting behavior, but they are also well known for engaging in synchronous nesting behavior, also termed "arribada", on tropical beaches like those in Mexico, Costa Rica, and India (Bernardo and Plotkin 2007, as cited in Fonseca, et. al., 2009). Arribada nesting behaviors are a key evolutionary strategy that promotes nest survival by overwhelming predators with sheer numbers, since satiated predators cannot continue to prey on other nests (Eckrich and Owens, 1995).

Despite its abundance and large range, *L. olivacea* was globally listed as a declining Vulnerable species by the International Union for the Conservation of Nature's (IUCN) Red List in 2008, making it a target for conservation efforts. One common strategy to recover sea turtle populations is to protect nests and increase recruitment (Eckert, et. al., 1999) by addressing common coastal threats like predation and erosion (Marcovaldi, 2001).

Due to arribada nesting being a direct evolutionary response to predation, understanding arribada behavior over time may be a crucial component to inform *L. olivacea* conservation management. Currently, it is known that *L. olivacea* arribada nesting in Ostional, Costa Rica is responsive to rainfall-related environmental cues like low salinity, higher relative humidity and lower nearshore current velocity (Bézy, et. al., 2020). Additionally, arribada abundance significantly increases during the rainy season in Ostional, though hatching success may decrease from increased erosion and microbial loads in nests affecting embryonic mortality (Valverde, et. al., 2012), underscoring the importance of understanding the link between weather and nesting patterns.

Data Analysis and Results

The data used in this analysis comes from the Costa Rica Ministry of Environment and comprises data measurements of nesting *L. olivacea* in Ostional nesting shores, spanning from 2019 to 2025. Data pertaining to the time and weather of nesting events were cleaned by hand in Microsoft Excel (Version 2510, Microsoft Corporation, 2025), then analyzed in Microsoft Excel and R Statistical Software (v4.5.2; R Core Team, 2025).

Nesting Event Distribution by Weather Type

Weather measurements in the raw dataset varied in description, but fell into four broad categories: clear, partly cloudy, cloudy, and stormy. These are the descriptions used in the following analysis. The frequency of all weather observations ($n = 1558$) was counted using different pivot tables for solitary events ($n = 902$) and arribada events ($n = 656$). These counts were used to create a 2x4 contingency table (see Table 1).

To determine if there was a relationship between nesting event type and weather conditions, the chi-square test of independence was used by considering the sum of the deviation of the observed frequencies with the expected value: $\chi^2 = \sum \frac{(O-E)^2}{E}$, where the expected value of an observation is derived from the contingency table as $E = \frac{row\ sum * column\ sum}{total\ sum}$.

According to this test, there is a highly significant relationship between our observed variables ($p < 0.00001$). To determine significant differences in solitary vs. arribada frequencies within weather types, a similar goodness-of-fit test was used for each weather type following the same formula for χ^2, where the expected value of an observation is derived from the contingency table as $E = \frac{row\ sum}{2}$. According to this test, solitary and arribada events showed significant differences in frequency in partly cloudy ($p < 0.01$), cloudy ($p < 0.00001$), and stormy ($p < 0.00001$) weather conditions (see Table 1).

A binomial logistic regression (Solitary = 0, Arribada = 1) was then applied to determine the extent to which arribada events were more or less likely to occur in non-clear conditions compared to clear conditions. This revealed that arribada nesting events are 38% less likely to occur in partly cloudy conditions (OR = 0.6214, 95% CI [0.420, 0.912]), 30% less likely to occur in cloudy conditions (OR = 0.6992, 95% CI [0.561, 0.872]), and 78% less likely to occur in rainy conditions (OR = 0.212, 95% CI [0.134, 0.339]).

Discussion

The results of the chi-square and logistic regression tests indicate that arribada events are less likely to occur during non-clear conditions compared to clear conditions. Because nesting event types are binary (arribada vs. solitary), this also implies that solitary events are more likely to occur during non-clear conditions. On the other hand, the chi-square goodness-of-fit test for clear conditions does not demonstrate that either nesting type is more likely to occur. Therefore, other environmental factors might serve as stronger cues to trigger arribada nesting among *L. olivacea* during clear conditions. This means that although clear weather conditions do not necessarily encourage arribada nesting, other weather conditions may deter these events. This is especially true for inclement weather conditions (rain, lightning storms) which showed the steepest reduction in arribada nesting.

This is particularly surprising because if the relationship between rainy weather and arribada nesting frequency were truly negative, it would be reasonable to expect fewer arribada nesting events during times when rain is more frequent. However, just the opposite is the case in Ostional, where significantly more arribada events are observed during the rainy season, which instead suggests a positive relationship. There are a number of reasons that could explain this conflict.

Bézy, et. al. (2020) highlighted that low salinity, high humidity, and low ocean current velocity conditions to be heavily associated with arribada events. This is further supported by Barik, et. al. (2014), who also found that low salinity and low energy oceanographic conditions are conducive to arribada nesting. One interesting speculation arribada nesting requires both rainy and clear weather conditions to happen sequentially, since this would produce all of these environmental cues described: recent storms create low salinity, high humidity conditions while present clear conditions ensure ocean current velocities are low. Bézy, et. al. (2020) also speculated that turtles might avoid nesting when short-term weather conditions makes the survival of eggs unlikely, including rainfall-induced erosion and increased microbial load (Valverde, et. al., 2012), further incentivizing turtles to emerge in clear conditions after a rainstorm. This would explain why arribada events happen more frequently during the rainy season, where rain-clear weather successions are more likely than the dry season, which would instead only produce clear conditions.

Climate change may affect weather patterns to induce hotter and drier conditions, and more extreme weather events (Castro-Vincenzi, 2024). This implies that if rainstorms are a significant cause for low salinity near shore surface water and increased relative humidity, then these signals will decrease as rainstorms become less frequent. So even though this analysis demonstrated that arribada nesting events are more likely during clear conditions, the reduction in arribada-signalling environmental cues may actually decrease arribada frequency. This information can be used to further push for climate resilience as a conservation management strategy, which may want to preserve arribada nesting behaviors as an anti-predation measure. While there doesn’t appear to be any one single cue that triggers arribada nesting, future studies may benefit from elucidating the potential for weather conditions to act as an environmental cue, especially when climate change threatens to disrupt historic weather patterns.

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Table 1

2x4 Contingency Table of Solitary and Arribada Nesting Event Frequency by Weather Type

	Nesting Event Type			
Weather Type	Solitary	Arribada	Total	*p*
Clear	301	299	600	9.35E-01
Partly Cloudy	81	50	131	6.76E-03
Cloudy/Foggy	404	282	686	3.19E-06
Storm/Rain/Lightning	116	25	141	1.81E-14
Total	902	656	1558	

Note. This table captures the frequency of observations across nesting and weather variables. The totals for nesting type is reflected across the bottom Total row, and the totals for weather type is reflected across the righthand Total column. The total number of observations is reflected in the bottom-right corner, at the intersection of the Total column and Total row.

* The p-values in the rightmost column reflect the results of individual goodness-of-fit tests. Note that $p > 0.05$ only for clear weather conditions.

Table 2

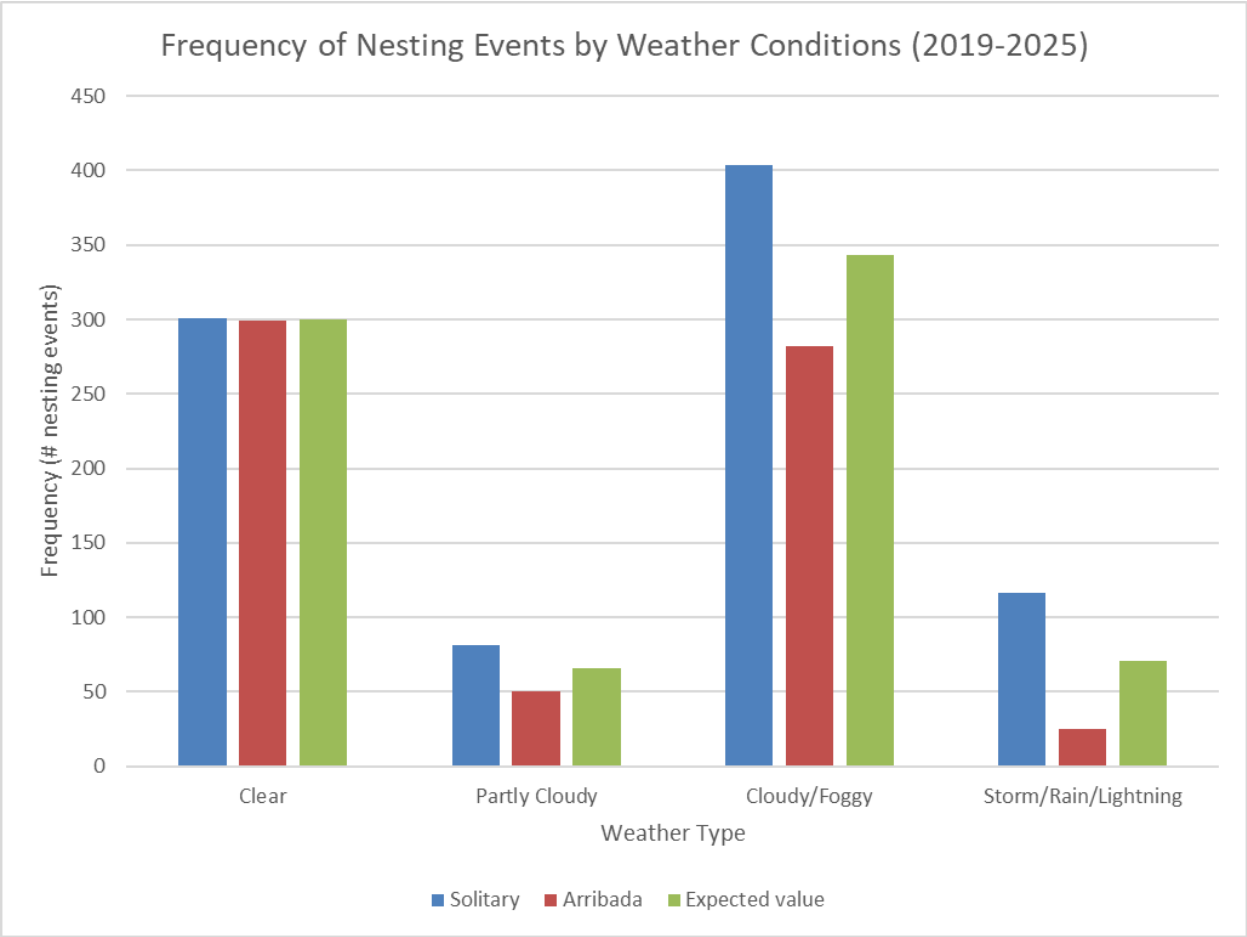
Odds Ratios of Arribada Events for Non-Clear Weather Events

	OR	2.5 %	97.5 %
(Intercept)	0.9933555	0.8463705	1.1658333
Cloudy	0.6992273	0.5605559	0.8716423
Rain	0.2169588	0.1341961	0.3385084
Partly cloudy	0.6214129	0.4199166	0.9121126

Note. This table captures the odds ratios (OR) for each non-clear weather event to have an arribada nesting event compared to clear weather conditions. The right two columns describe the bounds of the confidence interval (CI) at 95% confidence.

Figure 1

Bar Graph of Solitary and Arribada Nesting Frequency by Weather Type, with Expected Frequency



Note. This figure plots the frequency of Solitary vs. Arribada nesting events across four different weather groups. The expected value is set to be 50% of all nesting events within one weather group, intended for the chi-square goodness-of-fit test.